Five Raters Are Not a Distribution: What Learned Judges Can Learn from Human Disagreement, and Where to Spend the Annotation Budget

Anonymous

October 27, 2026

Abstract

Learned judges are increasingly trained on the full histogram of human ratings rather than on a single label, with the stated aim of learning how people disagree. We ask what part of such a histogram a text-conditioned judge can learn. We separate two properties that prior work conflates: whether the histogram's shape beyond its location (mode and mean) is *reproducible across raters*, and whether it is *predictable from the text*. On rating panels from 14 datasets — summarisation, offensiveness, politeness, helpfulness, sarcasm, valence, conversational safety, NLI, and seven software-engineering datasets including a 5,468-developer code-readability study — a learner-free split-half analysis finds the shape beyond mode and mean poorly reproducible on every ordinal panel: its split-half reliability stays between 0.04 and 0.41 with 4–8 raters and below 0.3 even with 40 raters on the dense panels, so the raw histogram never forecasts a new rater better than its own mode and mean. A cross-fitted decoding analysis finds that the frozen representations of three model families recover at most a few percent of the reproducible ordinal shape (upper 95 % bounds below 5 % on six of the eight ordinal panels with an informative interval — X distinguishable from zero and the G/X interval narrower than 20 points — and at most 10.2 % on all eight) but 17–26 % of the NLI shape. Learned judges behave accordingly: statistic-preserving replacement of the shape of 69–83 % of training targets changes LoRA judges by less than ±0.0011 RPS on two datasets, and alternative constructions of the per-item target change frozen-feature heads by at most a few thousandths of RPS in inconsistent directions, whereas discarding the distribution (one-hot targets) is worse on 10 of 11 panels. A rater–item decomposition of the panels with persistent rater identities shows that, beyond item location, rater-specific thresholds account for most of the held-out likelihood gain, while item-specific dispersion adds little. At a fixed number of ratings, frozen-feature judges improve with more items rather than more raters per item on most panels, and end-to-end LoRA judges trained on one rating per item match those trained on seven (two base-model families) and beat judges trained on seven ratings for a seventh of the items; the marginal value of further ratings on covered items is task-dependent. We release per-dataset disagreement reliability cards and the estimators.

1 Introduction

Human raters disagree, and a growing line of work trains learned judges — language models fine-tuned to score text — on the full distribution of human ratings rather than on a single aggregated label (Chen et al., 2025; Singh et al., 2025; Kohli, 2026; Benito-Santos and Ghajari, 2025). The appeal is clear: a histogram records which neighbouring levels received mass and how concentrated the ratings were, and a judge that reproduces it would "capture human disagreement" rather than average it away. The data behind that appeal are thin. In most datasets the histogram is computed from three to nine people. Five ratings of a summary's coherence are a sample from a population of possible raters, and a model trained to reproduce the sample exactly is, in part, trained to reproduce sampling noise.

This paper asks a question that the literature on learning from label variation (Uma et al., 2021; Plank, 2022) has not asked in this form: **which part of a few-rater histogram can a text-conditioned judge learn?** We decompose every histogram into its *location* (mode and mean) and the *shape beyond location* (the residual allocation of mass once mode and mean are fixed), and we measure two properties of the shape that are usually conflated:

1. **Reproducibility across raters.** If the same items were rated by a disjoint panel, would the shape come out the same? We estimate this learner-free with a split-half design: an exact identity, A1 = M − 2X, relates the forecasting value of the raw histogram to the magnitude M of its residual relative to an exact mode-and-mean comparator and the cross-panel agreement X of that residual; the ratio $\rho = X/M$ is a split-half reliability of the shape at a given number of raters, and the raw histogram beats the comparator for predicting a new rater exactly when $\rho > \frac{1}{2}$.
2. **Predictability from the text.** Even a reproducible shape is useless to a judge unless the text carries it. We fit a cross-fitted ridge decoder from a frozen LLM's representation to the panel-1 residual and score it on an independent panel-2 residual with a proper-score gain G, reported also as the fraction G/X of the reproducible shape, against a within-cell feature-permutation control and a positive control.

Findings. (i) *The shape beyond mode and mean is poorly reproducible on ordinal scales.* Its split-half reliability is 0.04–0.41 on eight panels with 4–8 raters per item (intervals include 0 on four), and with three raters the mode-and-mean cell determines the histogram outright (Crupi et al. (2025)'s code-summary benchmark, MLCQ, Roy et al.). On the dense ordinal panels it stays below 0.3 up to 40 raters (Buse–Weimer 0.22, DICES 0.25 and 0.15, Dorn's Java panel 0.29), so on no ordinal panel does the raw histogram forecast a new rater better than its own mode and mean. ChaosNLI is the contrast: the NLI shape's reliability exceeds $\frac{1}{2}$ from two to three raters and reaches 0.96–0.98 at 40. (ii) *What is reproducible is mostly not in the text.* A cross-fitted linear decoder on frozen features, scored against an independent rater panel, recovers a fraction G/X of the reproducible shape whose upper 95 % bound is 0.2–0.4 % on SummEval, ≤ 0.4 % on POPQUORN offensiveness, ≤ 2 % on EmoBank and DICES-350, ≤ 4.6 % on DICES-990, ≤ 5.4 % on Dorn and ≤ 10.2 % on HelpSteer2 (CSC's G is ≈ 0 but its X interval covers 0, so no ratio is reported); on the six panels run in all three families (Qwen2.5-7B, Qwen3-8B, OLMo-2-7B) the families agree; on ChaosNLI MNLI it recovers 17–18 % in all three families (intervals [0.11, 0.19]) and 26 % on SNLI (Table 2). (iii) *Learned judges do not use the shape.* Replacing the shape of 69–83 % of training targets while preserving each target's exact mode and mean changes LoRA judges by −0.0001 [−0.0011, +0.0011] RPS on SummEval and +0.0000 [−0.0009, +0.0009] on POPQUORN (position-level bootstrap; user-level [−0.0014, +0.0015]), whereas destroying the location as well costs +0.012 and +0.003 (POPQUORN user-level interval [−0.0004, +0.0065]); the same intervention on ChaosNLI costs 0.016–0.052 KL. Across eleven ordinal panels and three feature families, training frozen-feature heads on the location-only comparator or on reliability-shrunk shapes changes held-out RPS by at most 0.0012 with mixed signs; other smoothing constructions move it by −0.005 to +0.009 in panel-dependent directions; one-hot targets are worse on 10 of 11 panels. A user-held-out replication on POPQUORN, where no evaluator contributes to any target, gives the same signs. (iv) *Beyond location, rating structure is mostly rater-side.* On the four panels with persistent rater identities, rater-specific cut-points improve held-out likelihood by 0.05–0.16 nats per rating while item-specific dispersion adds −0.017 to +0.012. This is consistent with — though it does not by itself prove — the reading that, on these panels, the shape a judge is asked to learn is largely a property of who rated. (v) *Budget.* At a fixed number of ratings, frozen-feature heads improve with more items rather than more raters per item on four of the five ordinal panels with more than 900 items and on both ChaosNLI panels (SummEval is flat; panels with ≤ 210 training items reverse). End-to-end LoRA judges trained on one rating per item match judges trained on all seven on SummEval (Qwen2.5, same batch, 3 seeds: −0.0027 [−0.0061, +0.0014]; OLMo-2: +0.0007 [−0.0056, +0.0069]) and beat judges trained on 144 × 7 by 0.023 / 0.0095 RPS; on POPQUORN (an earlier two-seed block, not re-run in one batch) a second rating per item recovers about two thirds of the single-rating gap. A prediction we registered — that multi-rater items would pay off at fixed budget where the shape is text-predictable — was falsified on ChaosNLI; we therefore do not derive the budget result from the mechanism and report the task-dependence of the marginal value of extra ratings as an observation.

Contributions. (1) Two reliabilities for disagreement — across raters and across text — with learner-free, cross-fitted estimators. (2) Disagreement reliability cards for every panel, telling a dataset builder whether the shape of disagreement is reproducible at the number of raters they can afford. (3) A rater–item decomposition of the ordinal structure beyond location. (4) Fixed-budget evidence, with frozen-feature heads in three model families and end-to-end LoRA judges, that covering items matters more than adding raters for training. The donor-replacement audit (§3) is the evidence that current training does not use the shape; shrinkage of the histogram survives as a tool for forecasting a new rater, not as a training method.

What we do not claim. That disagreement is noise (part of it is reproducible); that the text contains no residual information (we test linear decodability from three frozen model families); a theoretical rater-count threshold (a two-term 1/a model was falsified); a causal role of rater thresholds or of label structure in the ordinal–nominal difference; or a universal budget rule (Zhang et al. (2021) find the opposite when a large

single-label base already exists; see §5).

2 b. Related work

Learning from label variation. Human label variation is increasingly treated as signal rather than noise (Aroyo & Welty 2015; Pavlick & Kwiatkowski 2019; Uma et al., 2021; Plank, 2022), and soft-label training is reported to improve calibration and robustness (Peterson et al., 2019; Collins et al., 2022; Singh et al., 2025). For learned judges, Chen et al. (2025) train on human rating distributions; Kohli (2026) shows that on ChaosNLI the item-specific information in soft labels saturates around ten annotators for KL. These studies compare training targets; none separates the part of the histogram that is reproducible across raters from the part that is predictable from the input, which is the distinction our results turn on.

Predicting disagreement. Disagreement is partly predictable from text: NLI label distributions can be predicted to a degree (Zhou et al., 2022), reward models detect diverging preferences with AUROC around 0.6 (Zhang et al., 2025), and demographic information helps predict who disagrees (Wan et al., 2023). Our decoding analysis gives the matching quantity for the shape beyond mode and mean, normalised by its reproducible part, and finds the NLI and ordinal cases on opposite sides.

Reliability of aggregates and rater effects. Reliability of aggregated annotations depends on the number of raters (Wong and Paritosh, 2022); rater severity and scale use are classical facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011). We use the split-half logic and the rater facet, but apply them to the shape beyond location and connect them to what a text-conditioned judge can learn.

Annotation budgets. Whether to label more items or relabel existing ones is an old question in crowdsourcing (Sheng et al., 2008; Lin et al., 2014); for hard-label classifiers, labelling many items once is preferable above a worker-quality threshold (Khetan et al., 2018). In NLP, Zhang et al. (2021) find the opposite on ChaosNLI: replacing a few thousand single-label items by a few hundred ten-label items reduces KL — in a regime where a large single-label base already exists and the evaluation targets the full distribution. Our budget results concern the complementary regime (every rating competes for items, ordinal quality ratings, judges trained only on the budgeted data), and on ChaosNLI we also observe a larger marginal value of a second rating than on ordinal panels; the two findings are consistent once the regime is stated.

3 Two reliabilities of a disagreement histogram

Setting and notation. Item i has ratings on K ordered levels from a panel of raters; the empirical histogram is h_i. For ordinal scales we score forecasts with the ranked probability score, $S(f, y) = \sum_k (F_f(k) - \mathbf{1}[y \le k])^2/(K-1)$, and write C for the cumulative operator; for nominal tasks (ChaosNLI) S is the Brier score and C the identity. The **location** of a histogram is its exact mode and mean (nominal: modal class and modal mass); the **shape beyond location** is what remains once these are fixed. To isolate it we use the *exact-T comparator* q_T (Appendix A): a cross-fitted forecast built from other items in the same exact mode-and-mean cell (KL-projected onto the cell when the cell is sparse), so that q_T has the same mode and mean as h_i and differs from it only in shape. The residual of a histogram is $r = C(h - q_T)$.

2.1 Reproducibility across raters (split-half reliability).

In each of R repetitions the ratings of every item are split at random into disjoint panels P_1 (size a) and P_2. Writing $r_1 = C(h_1 - q_T)$ and $r_2 = C(h_2 - q_T)$, with q_T computed from P_1, the forecasting value of the raw histogram over the comparator obeys the exact identity

$$A1 \;=\; \mathbb{E}\big[S(h_1, y) - S(q_T, y)\big]_{y \in P_2} \;=\; M - 2X, \qquad M = \|r_1\|^2/n,\; X = \langle r_1, r_2\rangle/n,$$

so $\rho = X/M$ is a split-half reliability of the shape and the raw histogram beats the location-only forecast exactly when $\rho > \frac{1}{2}$. M is the observed magnitude of the shape, X its agreement with an independent panel's shape; $M - X$ is sampling noise. ρ is specific to the number of raters a and, as a ratio, unstable when M is tiny; we report it with intervals and omit it where the residual is structurally empty (e.g. two or three ratings on a three-level scale). We also report the support of the donor-replacement intervention (the fraction of items whose histogram could be changed at all while preserving mode and mean). A smoother, response-based comparator that does not reproduce the noisy location gives larger values of ρ because it also credits location reliability; we

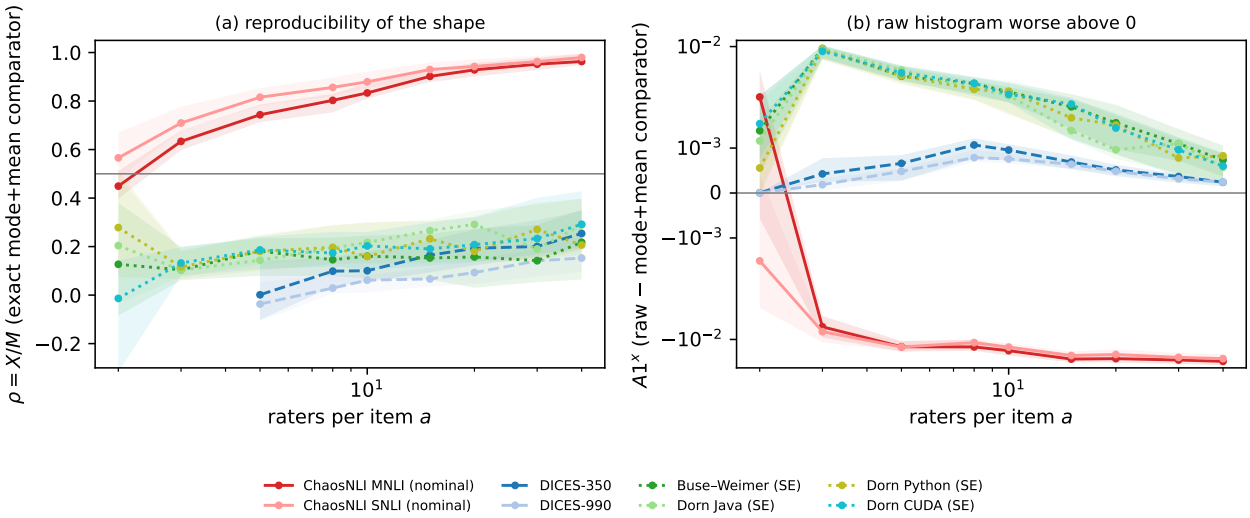


Figure 1: Reliability of the shape beyond exact mode and mean against the number of raters per item, dense panels. (a) Split-half reliability $\rho = X/M$ relative to the exact mode-and-mean comparator, with 95% item-cluster intervals; omitted where the residual is structurally empty or the ratio interval is wider than 1 (DICES with 2–5 ratings on a 3-level scale). The raw histogram beats the location-only forecast only above $\frac{1}{2}$. (b) $A1^x$: positive means the raw histogram forecasts a new rater worse than its own mode and mean.

use the exact comparator throughout. Intervals are item-cluster bootstraps in which all copies of a resampled item stay in one cross-fitting fold and the comparator is refitted in every replicate. Table 1 gives the resulting *reliability cards*; Figure 1 traces ρ and $A1^x$ against a on the dense panels.

2.2 Predictability from the text (decodability).

A judge sees the text, not the raters. Whether the reproducible shape is available to it is a separate question, which we answer with the representation the judges are built on: last-token hidden states of the frozen Qwen2.5-7B-Instruct (Qwen2.5-Coder for code panels) under the same rating prompt. A multi-output ridge decoder, group-cross-fitted over five item folds with a label-free PCA and a nested penalty, predicts r_1 from the features; it is scored on the *independent* panel r_2 by the covariance $V = \langle \hat{r}, r_2 \rangle / n$ and by the proper-score gain

$$G \;=\; \big(2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2\big)/n \;=\; \mathbb{E}_{y\in P_2}\big[S(q_T, y) - S(q_T + C^{-1}\hat{r},\, y)\big],$$

the exact reduction in score obtained by adding the decoded shape to the location-only forecast. Two controls accompany every estimate: a *within-cell permutation* of the features among items with the same exact mode and mean (which must drive G to zero or below), and a *positive control* that predicts the panel-1 mean with the same pipeline (which must be clearly positive, else the representation or the alignment is broken). Directions are symmetrised ($P_1 \leftrightarrow P_2$) and bootstrapped over item clusters. Because a single layer is a weak probe, we repeat the analysis with per-layer mean- and last-token features and greedy selection of the best layers (the probing recipe of Li et al., 2026).

2.3 Why the two can come apart.

ρ asks whether the shape is a property of the item's *rater population*; G asks whether it is a property of the item's *text*. A shape can be reproducible yet invisible to text — for instance when it is produced by how particular kinds of raters use the scale — and a text-predictable shape can be unreliable with few raters. Section 4 shows that the first case is the ordinal norm. Nothing in §2.1 or §2.2 trains a judge; both are measurements of the data and the representation, and both are released with the reliability cards so that a dataset builder can compute them before deciding how many raters to pay for.

Table 1: Disagreement reliability cards (exact mode-and-mean comparator). $\rho = X/M$: split-half reliability of the shape beyond mode and mean at a raters, 95% item-cluster interval; the raw histogram forecasts a new rater better than its own mode and mean only if $\rho > \frac{1}{2}$ ($A1^x < 0$). Support: share of items whose histogram can be changed at all with mode and mean fixed. Three-rater panels: the mode-and-mean cell determines the histogram (support 0). SE = software engineering.

panel	raters	a	ρ [95% CI]	$A1^x$	support
CSC sarcasm	6	5	0.04 [-0.06, 0.13]	+0.0050	0.72
EmoBank	5	4	0.41 [0.27, 0.55]	+0.0003	0.14
HelpSteer2	5	4	0.13 [0.04, 0.17]	+0.0036	0.45
POPQUORN off.	8	7	0.07 [-0.04, 0.25]	+0.0031	0.69
POPQUORN pol.	5	4	0.04 [-0.03, 0.12]	+0.0044	0.42
Scalabrino^{SE}	9	8	0.19 [-0.24, 0.55]	+0.0027	0.69
SummEval coh.	8	7	0.35 [0.24, 0.48]	+0.0016	0.87
SummEval rel.	8	7	0.40 [0.30, 0.54]	+0.0008	0.79
Buse–Weimer^{SE}	121	40	0.22 [0.06, 0.35]	+0.0007	0.89
DICES-350	104	40	0.25 [0.19, 0.35]	+0.0002	1.00
DICES-990	60	40	0.15 [0.10, 0.24]	+0.0002	1.00
Dorn CUDA^{SE}	160	40	0.29 [0.20, 0.43]	+0.0006	–
Dorn Java^{SE}	160	40	0.29 [0.16, 0.40]	+0.0006	–
Dorn Python^{SE}	160	40	0.21 [0.18, 0.40]	+0.0008	–
ChaosNLI MNLI	100	40	0.96 [0.95, 0.99]	−0.0186	1.00
ChaosNLI SNLI	100	40	0.98 [0.95, 1.00]	−0.0173	0.99
Crupi CA^{SE}	3	2	– (no shape)	–	0.00
Crupi conc.^{SE}	3	2	– (no shape)	–	0.00
Crupi flu.^{SE}	3	2	– (no shape)	–	0.00
MLCQ^{SE}	3	2	– (no shape)	–	0.00
Roy CA^{SE}	3	2	– (no shape)	–	0.00

4 What learned judges use: the audit, condensed

If a judge learned the shape beyond location, destroying that shape in its training targets would hurt it. A donor-replacement audit (Appendix A gives the full design and tables) does exactly this with certified dose: for 69–83 % of training items the histogram is replaced by another item's histogram projected onto the recipient's exact mode and mean (mean TV 0.30–0.35 among changed items), under two learners. **End-to-end LoRA judges** (Qwen2.5-7B-Instruct, the distributional objective of Chen et al., 2025; three optimizer–assignment realisations × three held-out-rater folds) change by −0.0001 [−0.0011, +0.0011] RPS on SummEval coherence and +0.0000 [−0.0009, +0.0009] on POPQUORN offensiveness, whereas destroying the location as well (mode-only replacement) costs +0.0120 [+0.0086, +0.0155] and +0.0029 [+0.0009, +0.0050] with every seed-by-fold cell positive (POPQUORN intervals resample rating positions; resampling users instead widens them to [−0.0014, +0.0015] and [−0.0004, +0.0065]). **Frozen-feature heads** on ten ordinal panels move by −0.0024 to +0.0006. On the nominal ChaosNLI panels the same kind of intervention — scrambling the allocation of mass beyond the modal class while preserving the modal mass — raises test KL by 0.016–0.052 at every annotation count.

A stronger version of the same test asks whether *any* construction of the per-item target helps. We trained the heads on eleven ordinal panels with the raw histogram, the location-only comparator, reliability-shrunk shapes and locations, Dirichlet smoothing toward the mode prototype, label smoothing toward uniform and toward the marginal, and a cross-item smoothed comparator, thirty configurations each. The location-only comparator and reliability-shrunk shapes change held-out RPS by at most 0.0012 with mixed signs; the other smoothing constructions move it by −0.005 to +0.009 with signs that differ across panels; the one-hot target is worse on 10 of 11 panels (up to +0.010; Scalabrino −0.0013). The learner pools location across items on its own and shows no sensitivity to the per-item shape — what §2.2 leads one to expect for a shape that is not linearly decodable from its features. Learned judges on ordinal scales learn location and calibration; on NLI they also learn shape, because there the shape is in the text.

5 Where the ordinal shape comes from: a rater–item decomposition

The split-half analysis says the few-rater shape is mostly noise but not entirely ($\rho > 0$ with an interval excluding 0 on half of the small panels and on all dense panels at 40 raters); the decoding analysis says the reproducible

Table 2: Decodability of the residual beyond exact mode and mean from frozen representations of three model families: decoded proper-score gain on an independent rater panel as a percentage of the panel's reproducible shape, $100\,G/X$ (95% item-cluster bootstrap); positive control in parentheses (Spearman of the decoded vs held-out mean; point estimate). "X n.s.": the reproducible shape itself has an interval covering 0, so no ratio is reported.

panel	Qwen2.5-7B	Qwen3-8B	OLMo-2-7B
Buse–Weimer^{SE}	+6.6 [-7.0, +26.0] (0.54)	–	–
CSC sarcasm	X n.s. (0.44)	X n.s. (0.51)	X n.s. (0.52)
DICES-350	-2.4 [-18.5, +1.4] (0.47)	–	–
DICES-990	-0.1 [-24.6, +3.2] (0.71)	-2.9 [-13.8, +4.6] (0.72)	-3.3 [-7.9, +2.9] (0.75)
Dorn (all)^{SE}	+3.2 [-2.7, +5.4] (0.67)	–	–
EmoBank	+0.9 [-0.8, +2.0] (0.48)	+0.6 [-0.8, +1.8] (0.48)	+1.1 [-1.0, +1.7] (0.49)
HelpSteer2	+2.5 [-5.0, +10.2] (0.36)	+4.1 [-3.9, +9.8] (0.39)	+0.9 [-7.6, +7.3] (0.27)
JetBrains naming^{SE}	+1.1 [-37.3, +2.9] (0.30)	–	–
POPQUORN off.	-1.6 [-4.9, +0.0] (0.65)	-0.8 [-4.2, +0.4] (0.65)	-2.0 [-4.9, +0.1] (0.68)
POPQUORN pol.	X n.s. (0.73)	–	–
Scalabrino^{SE}	X n.s. (0.41)	–	–
SummEval coh.	-0.1 [-1.0, +0.2] (0.41)	+0.1 [-0.9, +0.3] (0.36)	-0.2 [-1.5, +0.2] (0.27)
SummEval rel.	-0.1 [-2.5, +0.4] (0.33)	–	–
ChaosNLI MNLI	+16.9 [+11.2, +17.6] (0.29)	+18.2 [+13.3, +19.1] (0.32)	+17.7 [+12.4, +18.3] (0.24)
ChaosNLI SNLI	+26.0 [+20.4, +27.0] (0.41)	–	–

part is not in the text. What is it, then? We fit three ordered-probit models to the rating matrices of the panels with persistent rater identities (Buse–Weimer, DICES-350, Scalabrino, the SummEval experts) and location-plus-dispersion models to the position panels: $Y^*_{ir} = \theta_i + \sigma_i\,\varepsilon_{ir}$ with cut-points c_r; **M0** item location with common cut-points, **M1** adding rater-specific cut-points (harshness), **M2** adding item-specific dispersion σ_i (the inverse of IRT discrimination). Held-out log-likelihood over random cells, five folds, with the dispersion penalty chosen on held-out data.

Rater thresholds carry the structure: M1 improves held-out likelihood by +0.111 (BW), +0.146 (DICES-350), +0.156 (Scalabrino), +0.104 / +0.047 (SummEval coherence / relevance) nats per rating; item dispersion adds −0.017 (Scalabrino) to +0.012 and over-fits on the 5–7-rating position panels. Simulating rating matrices from the fitted models and re-running the split-half analysis closes the loop: at the split used for this check, the observed exact-T residual agreement on the dense ordinal panels is X ≈ 0 (BW −0.0001, DICES-350 +0.00003, DICES-990 +0.00002) and every model reproduces it; with 40 raters a small reproducible shape does remain ($\rho = 0.22$ [0.06, 0.35] on BW, 0.25 [0.19, 0.35] on DICES-350; Table 1), and we have not tested whether the threshold model reproduces it at that size. SummEval retains a small residual (X = +0.0018 / +0.0016) that neither thresholds nor one-dimensional dispersion reproduce, and it is precisely this residual that the decoder of §2.2 cannot read from the text. The mechanism is the textbook one of many-facet Rasch measurement (Linacre, 1989; Eckes, 2011), and we present it as measurement, not as a claim about raters' psychology; its consequence for judges is new: on the panels with persistent rater identities, most of the structure beyond location that a text-conditioned model is asked to learn is associated with who rated — information that is not in the item text.

The annotation side confirms the same picture from a different angle. For forecasting a new rater from an item's own ratings, shrinking both location and shape toward the mode prototype by their split-half reliabilities improves RPS by −0.0048 [−0.0052, −0.0045] over the raw histogram (eight panels, inverse-variance pooled; panels are not independent), a cross-item smoothed comparator by −0.0117 [−0.0123, −0.0112]; shrinking the *shape* alone by its reliability $\lambda = X/M$ beats the raw histogram (−0.0020 [−0.0022, −0.0019]) but is indistinguishable from the location-only comparator (pooled −0.0000 [−0.0000, +0.0000]) on every panel except SummEval and EmoBank, where $\hat{\lambda} \approx 0.4$ (results/shrinkage/SUMMARY.md, corrected cluster bootstrap). For forecasting a new rater, what a few-rater histogram contributes beyond its location is small.

6 Spending the annotation budget

If the shape is unlearnable and the location is what the judge learns, then every rating after the first on the same item buys at most a less noisy location, while a rating on a new item buys a new point for the regression

Table 3: Frozen-feature heads at a fixed rating budget (Qwen2.5-7B features; soft-label head): test RPS (ordinal; ChaosNLI: KL to a held-out 50-rater reference) for items × ratings per item at the same total, mean over 30 configurations (5 seeds × 3 item splits × 2 rater splits). Lower is better; best per row in bold.

panel	ratings	items × ratings: score
Buse–Weimer^{SE}	60	60×1: 0.187, **30×2: 0.171**
DICES-350	210	210×1: 0.272, **105×2: 0.241**, 52×4: 0.242, 26×8: 0.248
DICES-990	594	**594×1: 0.203**, 297×2: 0.205, 148×4: 0.207, 74×8: 0.213, 40×15: 0.219
Dorn Java^{SE}	72	72×1: 0.163, **36×2: 0.163**
EmoBank	1800	**1800×1: 0.088**, 900×2: 0.089, 450×4: 0.090
HelpSteer2	2870	**2870×1: 0.201**, 1436×2: 0.202, 718×4: 0.204
POPQUORN off.	819	**819×1: 0.133**, 410×2: 0.136, 273×3: 0.137, 136×6: 0.140
SummEval coh.	960	960×1: 0.172, **480×2: 0.170**, 320×3: 0.170
ChaosNLI MNLI	959	**959×1: 0.226**, 480×2: 0.234, 240×4: 0.260, 120×8: 0.274, 60×16: 0.298, 30×32: 0.323
ChaosNLI SNLI	908	**908×1: 0.244**, 454×2: 0.258, 227×4: 0.298, 114×8: 0.353, 57×16: 0.408, 28×32: 0.445

Table 4: End-to-end LoRA judges (distributional objective; Qwen2.5-7B and, for SummEval, OLMo-2-7B) at fixed rating budgets: test RPS minus the full-data judge, 95% crossed bootstrap over optimizer seeds and test groups.

panel	design (items × ratings)	ratings	RPS	− full data
SummEval (Qwen2.5, same batch, 3 seeds)	960 × 1	960	0.1664	−0.0027 [−0.0061, +0.0014]
SummEval (Qwen2.5, same batch, 3 seeds)	480 × 2	960	0.1651	−0.0040 [−0.0084, −0.0000]
SummEval (Qwen2.5, same batch, 3 seeds)	144 × 7	1008	0.1897	+0.0206 [+0.0163, +0.0247]
SummEval (Qwen2.5, same batch, 3 seeds)	960 × 7	6720	0.1691	–
POPQUORN (Qwen2.5)	819 × 1	819	0.1294	+0.0077 [+0.0046, +0.0110]
POPQUORN (Qwen2.5)	117 × 7	819	0.1322	+0.0105 [+0.0065, +0.0151]
POPQUORN (Qwen2.5)	410 × 2	820	0.1339	+0.0122 [+0.0083, +0.0167]
POPQUORN (Qwen2.5)	205 × 4	820	0.1352	+0.0135 [+0.0096, +0.0172]
POPQUORN (Qwen2.5)	819 × 2	1638	0.1241	+0.0024 [+0.0002, +0.0046]
POPQUORN (Qwen2.5)	234 × 7	1638	0.1277	+0.0061 [+0.0030, +0.0092]
POPQUORN (Qwen2.5)	410 × 4	1640	0.1282	+0.0065 [+0.0036, +0.0097]
POPQUORN (Qwen2.5)	819 × 7	5733	0.1217	–
SummEval (OLMo-2)	960 × 1	960	0.1801	+0.0007 [−0.0056, +0.0069]
SummEval (OLMo-2)	480 × 2	960	0.1778	−0.0016 [−0.0062, +0.0023]
SummEval (OLMo-2)	144 × 7	1008	0.1896	+0.0102 [+0.0049, +0.0156]
SummEval (OLMo-2)	960 × 7	6720	0.1794	–

the judge is implicitly fitting. We tested this directly by holding the total number of ratings fixed and trading items for raters.

Frozen-feature heads (eight ordinal and two nominal panels, thirty configurations each; Table 3; CSC is excluded because its 32 conversation contexts leave fewer than the design's minimum of 20 training groups). On every panel with more than about 300 training items except SummEval, more items with fewer ratings each wins monotonically: on DICES-990 at a budget of 594 ratings, 594 items × 1 rating gives RPS 0.203, 297 × 2 0.205, 148 × 4 0.207, 74 × 8 0.213 and 40 × 15 0.219; POPQUORN, HelpSteer2 and EmoBank follow the same order, SummEval is flat. The direction reverses only on small item sets (Buse–Weimer with 60 training items, DICES-350 with 210), where a single rating per item is too noisy a target. The same items-first order holds on ChaosNLI (959 × 1: KL 0.226; 30 × 32: 0.323) — a result that falsified a prediction we had registered, namely that multi-rater items would pay off at fixed budget where the shape is text-predictable. They do not; where tasks differ is the *marginal* value of a second rating on already-covered items (ChaosNLI −15 %, ordinal heads −1 to −3 %), which we report as an observation.

End-to-end LoRA judges (Table 4). On SummEval coherence, with the Chen et al. objective and the three held-out-expert folds of §3, re-run in one batch with three optimizer seeds × three folds (SHRINKAGE_PROTOCOL Amendment 6; results/lora_budget_v2/), a judge trained on 960 items × 1 randomly chosen rating (960 ratings) matches the judge trained on all 960 × 7 = 6,720 ratings (−0.0027 [−0.0061, +0.0014], crossed bootstrap over seeds and test documents); 480 × 2 is equally good (−0.0040 [−0.0084, −0.0000]; vs 960 × 1: −0.0013 [−0.0073, +0.0039]), while 144 × 7 is worse than 960 × 1 by +0.0233 [+0.0176, +0.0285] — both prespecified readings ("one rating per item matches the full data", "items first") hold. An earlier two-seed block

whose full-data reference came from a different batch had suggested that 960×1 *beats* the full data (-0.0061); the same-batch re-run does not support that stronger claim. (The earlier block also gave 240×4: $+0.0162$ and 960×2: -0.0064 vs full; these designs were not re-run.) On POPQUORN offensiveness (an earlier block: two seeds, full-data reference from the audit batch, not re-run in one batch) the ordering within each budget is again items-first (819×1 best at 820 ratings, 819×2 best at 1,638), but here extra ratings per item do help: 819×1 is $+0.0077$ [$+0.0046$, $+0.0110$] worse than the full 5,733-rating data and 819×2 recovers two thirds of that gap. The decisive part replicates on a non-Qwen base: with OLMo-2-7B on SummEval, 960×1 matches the full data ($+0.0007$ [-0.0056, $+0.0069$]) and 144×7 is worse than 960×1 by $+0.0095$ [$+0.0022$, $+0.0160$] (results/lora_budget_olmo/summary_vs_a1_F1.0.md); one versus two ratings per item is a tie on both bases. Single-rating training behaves like stochastic label sampling (Gheibi and Ghazizadeh, 2026) and is consistent with the many-once-over-few-many results for hard-label classifiers (Khetan et al., 2018; Lin et al., 2014); our contribution is the demonstration with distributional targets and end-to-end LLM judges, and the explanation of §§2–4 for why the raters' side of the budget buys so little.

Rule. For training a judge on an ordinal rating task, cover the items first; a second rating per item is worth between nothing (SummEval) and about two thirds of the single-rating gap (POPQUORN), and further ratings less. Raters belong where a reliable reference distribution is itself the deliverable — evaluation sets, and tasks such as NLI where the shape is in the text.

7 Software-engineering panels

Readability, summary quality and code-smell severity are rated on the same few-rater scales, and the same questions arise for code judges. Dorn's readability study (360 snippets in Java, Python and CUDA; 161–267 developers per snippet) gives the densest ordinal panel we know of; its reliability curve matches Buse–Weimer and DICES: the shape beyond mode and mean reaches a split-half reliability of only 0.2–0.3 even with 40 developers per snippet (Figure 1). The LLM-judge validation benchmarks of the field have three raters per item — Crupi et al.'s code summaries, MLCQ's smell severities, Roy et al.'s summaries — and with three raters the mode-and-mean cell determines the histogram: there is no shape to learn or to evaluate (support 0 %). Scalabrino's nine-rater panel sits in between (Table 1). Decoding from frozen features recovers none of it at a precision that matters: G/X is $+0.03$ [-0.03, $+0.05$] on Dorn's readability panel (360 snippets), negative on Scalabrino ($G < 0$, a small over-fitting penalty on 124 snippets; its X interval covers 0, so no ratio is reported), and uninformative on Buse–Weimer ($+0.07$ [-0.07, $+0.26$], 100 snippets) and JetBrains naming ($+0.01$ [-0.37, $+0.03$], 120 identifiers) (results/decoding_v2/). For SE human studies the practical reading is that three to five raters per artifact fix its location and nothing else, and that adding raters to a few artifacts does not make the shape of developer disagreement reproducible on these readability and quality scales.

8 Limitations

Base models: all frozen-feature results (decoding, target construction, fixed budget) replicate on Qwen3-8B and on the non-Qwen OLMo-2-7B (Table 2: ordinal G/X intervals include 0 on all six three-family panels with upper bounds ≤ 10.2 %, ChaosNLI MNLI G/X 0.17 / 0.18 / 0.18 in Qwen2.5 / Qwen3 / OLMo-2; budget orderings agree except HelpSteer2 on OLMo-2, where 1436×2 edges out 2870×1 by 0.0007; target-construction contrasts agree in sign except one-hot, whose cost differs by up to 0.009 across families), and the end-to-end budget result replicates with an OLMo-2-7B LoRA judge on SummEval (the audit's LoRA blocks remain Qwen2.5-only); frozen-feature heads for breadth, end-to-end judges on two datasets; linear decoders — we re-tested with per-layer features and greedy layer selection, not with non-linear or fine-tuned probes; SummEval's small unexplained residual; POPQUORN's position folds (a user-held-out check with real user ids, evaluators never in any target, confirms the four conclusions on offensiveness and politeness: mode+mean-preserving replacement -0.00004 / $+0.00024$ (the latter three SEMs above zero, but two orders of magnitude below the mode-only cost), mode-only $+0.0022$ / $+0.0029$, one-hot $+0.0042$ / $+0.0050$, 4 ratings on a quarter of the items $+0.0042$ / $+0.0030$ worse than 1 rating on all); one registered prediction falsified and reported; effect sizes in the third decimal of RPS, to be read against the mode-only penalty of $+0.012$ and the budget effects of $+0.02$. We make no claim that disagreement is noise, that text contains no residual information, that a theoretical rater count exists, or that label structure causes the ordinal–nominal difference.

References

Benito-Santos and Ghajari. Beyond averages: Learning with annotator disagreement in STS. In *Proceedings of EMNLP*, 2025.

Luyu Chen, Zeyu Zhang, Haoran Tan, Quanyu Dai, Hao Yang, Zhenhua Dong, and Xu Chen. Beyond single-point judgment: Distribution alignment for LLM-as-a-judge. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2505.12301.

Katherine M. Collins, Umang Bhatt, and Adrian Weller. Eliciting and learning with soft labels from every annotator. In *Proceedings of HCOMP*, 2022.

Giuseppe Crupi, Rosalia Tufano, Alejandro Velasco, Antonio Mastropaolo, Denys Poshyvanyk, and Gabriele Bavota. On the effectiveness of LLM-as-a-judge for code generation and summarization. *arXiv preprint arXiv:2507.16587*, 2025.

Thomas Eckes. *Introduction to Many-Facet Rasch Measurement*. Peter Lang, 2011.

Mirerfan Gheibi and Gashin Ghazizadeh. Same target, different basins: Hard vs. soft labels for annotator distributions. *arXiv preprint arXiv:2605.20642*, 2026.

Ashish Khetan, Zachary C. Lipton, and Animashree Anandkumar. Learning from noisy singly-labeled data. In *International Conference on Learning Representations (ICLR)*, 2018. arXiv:1712.04577.

Guneet Kohli. Metric-dependent annotation saturation for learning from label distributions. *arXiv preprint arXiv:2605.29797*, 2026.

Zhuochun Li et al. Rethinking LLM-as-a-judge: Representation-as-a-judge with small language models via semantic capacity asymmetry. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2601.22588.

Christopher H. Lin, Mausam, and Daniel S. Weld. To re(label), or not to re(label). In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*, 2014.

John Michael Linacre. *Many-Facet Rasch Measurement*. MESA Press, 1989.

Joshua C. Peterson, Ruairidh M. Battleday, Thomas L. Griffiths, and Olga Russakovsky. Human uncertainty makes classification more robust. In *Proceedings of ICCV*, 2019.

Barbara Plank. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of EMNLP*, 2022.

Victor S. Sheng, Foster Provost, and Panagiotis G. Ipeirotis. Get another label? improving data quality and data mining using multiple, noisy labelers. In *Proceedings of KDD*, 2008.

Agamdeep Singh, Ashish Tiwari, Hosein Hasanbeig, and Priyanshu Gupta. Distributions in, distributions out: The case for soft-label training. *arXiv preprint arXiv:2511.14117*, 2025.

Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72, 2021.

Ruyuan Wan, Jaehyung Kim, and Dongyeop Kang. Everyone's voice matters: Quantifying annotation disagreement using demographic information. In *Proceedings of AAAI*, 2023.

Ka Wong and Praveen Paritosh. k-rater reliability: The correct unit of reliability for aggregated human annotations. In *Proceedings of ACL (Short Papers)*, 2022.

Michael JQ Zhang et al. Diverging preferences: When do annotators disagree and do models know? In *Proceedings of ICML*, 2025.

Shujian Zhang, Chengyue Gong, and Eunsol Choi. Learning with different amounts of annotation: From zero to many labels. In *Proceedings of EMNLP*, 2021.

Xiang Zhou, Yixin Nie, and Mohit Bansal. Distributed NLI: Learning to predict human opinion distributions for language reasoning. In *Findings of ACL*, 2022.